

JEE-Main-05-04-2026 (Memory Based)
[EVENING SHIFT]
Physics

Question: Escape velocity on earth is $v_e = \sqrt{2gR}$. A planet of half the radius of earth will have density equal to that of earth, the escape speed shall be $\frac{v_e}{N}$ then N is

Answer: (N = 2)

Question: Find dimensions of $\sqrt{\frac{Gh}{C^5}}$

Options:

- (a) ML^2T^{-1}
- (b) $M^0L^0T^1$
- (c) M^0LT^{-1}
- (d) $M^0L^0T^{-1}$

Answer: (b)

Question: Velocity versus time graph Find magnitude of acceleration of the particle at $t = 5$ sec



Options:

- (a) $u_0/10$
- (b) $u_0/20$
- (c) $u_0/5$
- (d) $u_0/4$

Answer: (b)

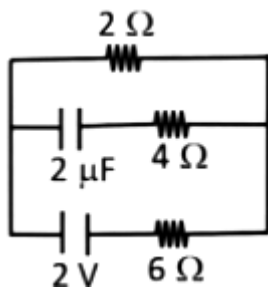
Question: 8 Hg drops coalesce to form a new drop. Ratio of final surface energy of simple drop to total surface energy of 8 drops is

Options:

- (a) $1/4$
- (b) $1/8$
- (c) $1/2$
- (d) 1

Answer: (c)

Question: In circuit below, find voltage across capacitor in steady state



Options:

- (a) 1V
- (b) 0.5 V
- (c) $\frac{3}{2}$ V
- (d) 4V

Answer: (b)

Question: A particle of mass m moves from height $2R$ above earth surface to surface of earth. Find change in P.E. (R is radius of earth)

Options:

- (a) $\frac{-2}{3} \frac{GMm}{R}$
- (b) $\frac{-GMm}{3R}$
- (c) $\frac{R}{GMm}$
- (d) $\frac{4R}{GMm}$

Answer: (a)

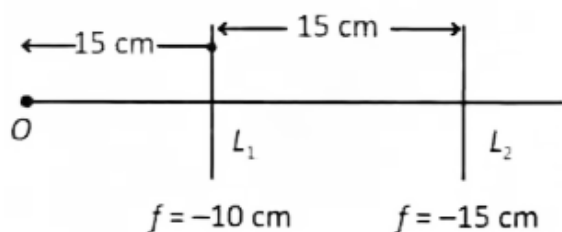
Question: Magnetic field $B = (2t^2 + 2t + 4)$ T is passing perpendicularly through a coil of radius $r = 20$ cm. Resistance of the coil is $R = 10$ Q. Find current through the coil at $t = 3$ sec.

Options:

- (a) $I = 8 \pi \times 10^{-2}$ A
- (b) $I = 3.2 \pi \times 10^{-2}$ A
- (c) $I = 7.2 \pi \times 10^{-2}$ A
- (d) $I = 5.6 \pi \times 10^{-2}$ A

Answer: (d)

Question: Two concave lens are placed at separation of 15 cm, find final image of object O.

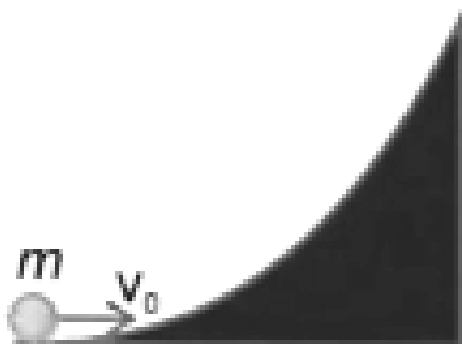


Options:

- (a) $\frac{35}{2}$ cm from L_2 & virtual
 (b) $\frac{17}{2}$ cm from L_1 & real
 (c) $\frac{34}{4}$ cm from L_2 & virtual
 (d) $\frac{17}{2}$ cm from L_2 & real

Answer: (c)

Question: Which of the following objects given in options can reach maximum height in situation shown in figure below : (given that there is no slipping)



Options:

- (a) Solid cylinder
 (b) Solid sphere
 (c) Disc
 (d) Ring

Answer: (d)

Question: An ideal gas undergoes process whose equation is $PT^3 = \text{constant}$ what would be polytropic constant for this

Options:

- (a) $1/2$
 (b) $3/4$
 (c) $2/3$
 (d) 0.4

Answer: (b)

Question: Assertion (A): EM wave exert pressure on surface on which it falls.

Reason (R): Rest mass of photons is zero.

Options:

- (a) Both A & R are correct & R is correct explanation of A
- (b) Both A & R are correct but R is not correct explanation of A
- (c) A is correct but R is false
- (d) A is false but R is correct

Answer: (b)

Question: An electron is moving with speed $0.8c$ in a medium. de-Broglie wavelength of electron in medium is (c is speed of light)

Options:

- (a) $1.54 \times 10^{-15} \text{ m}$
- (b) $3 \times 10^{-12} \text{ m}$
- (c) $2.34 \times 10^{-27} \text{ m}$
- (d) $3.5 \times 10^{-15} \text{ m}$

Answer: (b)

Question: If velocity of a particle is decreased by 20% then new de-Broglie wavelength is $\alpha \lambda_0$, where λ_0 is initial wavelength.

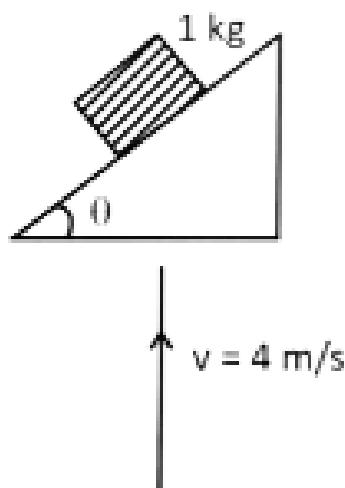
Find the value of α

Options:

- (a) $\alpha = 1.50$
- (b) $\alpha = 0.5$
- (c) $\alpha = 1.25$
- (d) $\alpha = 0.75$

Answer: (c)

Question: A block of mass 1 kg rests on an inclined plane. If whole system is moving with velocity 4 m/s (upward). Calculate the work done by friction in $t = 2 \text{ sec}$.



Options:

- (a) $\Delta w = 40 \sin 2\theta$
- (b) $\Delta w = 80 \sin^2 \theta$
- (c) $\Delta w = 80 \sin 2\theta$
- (d) $\Delta w = 40 \sin^2 \theta$

Answer: (b)

Question: A uniformly angular accelerated wheel travels θ_1 in the first 2 seconds and θ_2 in next θ_2 seconds. Initial angular speed is zero then $\frac{\theta_2}{\theta_1}$

Options:

- (a) 4
- (b) 3
- (c) 5
- (d) 6

Answer: (b)

Question: If a cell is connected to $20\ \Omega$ resistance then current in circuit is 0.25A . But if resistance of $2\ \Omega$ is connected across the cell then current is 2A . Find internal resistance of the cell.

Options:

- (a) $\frac{4}{7}\ \Omega$
- (b) $\frac{2}{3}\ \Omega$
- (c) $1\ \Omega$
- (d) $2\ \Omega$

Answer: (a)

Question: Consider ${}^6\text{C}^{12}$ Nuclei

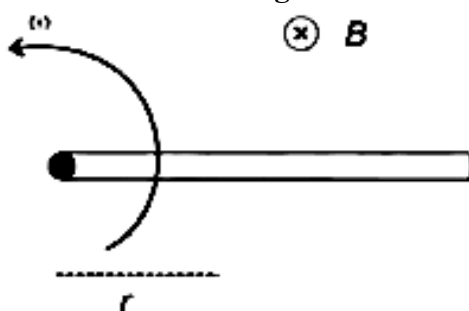
Given $m_p = 1.007276\text{ amu}$; $m_n = 1.008664\text{ amu}$. Find binding energy for nucleon. (in MeV/nucleon)

Options:

- (a) 7.42
- (b) 10.4
- (c) 2.20
- (d) 12.3

Answer: (a)

Question: Spatial magnetic field is given as $B = B_0 e^{-\lambda r}$, where r is distance from hinged end of the rod of length L . Emf induced across ends of rod for angular velocity of ω is



Options:

- (a) $\omega B_0 \left(\frac{1}{\lambda^2} - \frac{e^{-\lambda L}}{\lambda^2} \right)$

(b) $B_0 \left(\frac{e^{-\lambda l}}{\lambda^2} + \frac{l}{\lambda} e^{-\lambda l} \right)$

(c) $\omega B_0 \left(\frac{1}{\lambda^2} - \frac{e^{-\lambda l}}{\lambda} - e^{-\lambda l} \right)$

(d) $B_0 \left(\frac{e^{-\lambda t}}{\lambda} \right)$

Answer: (c)



JEE-Main-05-04-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: Given below are two statements:

Statement I: Glucose is found to exist in two different anomeric forms α and β .

Statement II: In open chain structure, C_3 , C_4 , C_5 of glucose & fructose both have same orientation.

Options:

- (a) Both statement I and statement II are correct
- (b) Statement I is correct but statement II is incorrect
- (c) Statement I is incorrect but statement II is correct
- (d) Both statement I and statement II are incorrect

Answer: (a)

Solution: (a) Both statement I and statement II are correct

Question: Given below are two statements:

Statement I: The shape of ICl_3 is square planar

Statement II: The shape of $ClBr_2^-$ is pyramidal

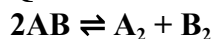
Options:

- (a) Both statement I and statement II are correct
- (b) Statement I is correct but statement II is incorrect
- (c) Statement I is incorrect but statement II is correct
- (d) Both statement I and statement II are incorrect

Answer: (d)

Solution: (d) Both statement I and statement II are incorrect

Question: Consider the following equilibrium-



If equilibrium pressure is P and degree of dissociation is α , then K_p is

Options:

- (a) $\frac{\alpha^2}{4(1-\alpha)^2}$
- (b) $\frac{p\alpha^2}{4(1-\alpha)}$
- (c) $\frac{p\alpha}{(1-\alpha)}$
- (d) $\frac{p\alpha}{4(1-\alpha)}$

Answer: (a)

Solution: (a) $\frac{\alpha^2}{4(1-\alpha)^2}$

Question: Which compound have total molecular mass of 72 with three primary carbons.

Options:

- (a) n-pentane
- (b) 2, 2-Dimethylbutane
- (c) 2, 2-Dimethylbutane
- (d) 2-methylbutane

Answer: (d)

Solution: (d) 2-methylbutane

Question: The order of acidic strength in following compounds is:

A. Phenol

B. p-nitrophenol

C. Ethanol

D. p-methoxyphenol

Options:

- (a) $B > A > D > C$
- (b) $C > B > A > D$
- (c) $B > D > A > C$
- (d) $B > A > C > D$

Answer: (a)

Solution: (a) $B > A > D > C$

Question: Consider the following electrochemical cell

$\text{Zn} | \text{Zn}^{2+} (\text{aq}) || \text{Ag}^+ (\text{aq}) | \text{Ag}$
(1 M)

The E_{cell} was found to be 1.6 V.

The value of $\log[\text{Ag}^+]$ is

Given $E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76\text{V}$, $E_{\text{Ag}^+/\text{Ag}}^{\circ} = -0.8\text{V}$

Options:

- (a) 1/3
- (b) 2/3
- (c) 3/2
- (d) 4/3

Answer: (b)

Solution: (b) 2/3

Question: An aqueous solution of a compound 'X' is acidified with H_2SO_4 and SO_2 is passed through it, turning the solution green. The compound 'X' is:

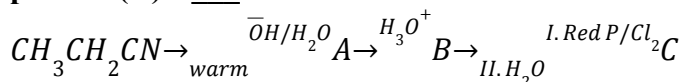
Options:

- (a) KMnO_4
- (b) $\text{Pb}(\text{CH}_3\text{COO})_2$
- (c) $\text{K}_2\text{Cr}_2\text{O}_7$
- (d) $\text{Fe}_2(\text{SO}_4)_3$

Answer: (c)

Solution: (c) $\text{K}_2\text{Cr}_2\text{O}_7$

Question: Consider the following sequence of reactions and determine the final major product(C) is ____



Options:

- (a) $\text{CH}_3\text{CH}_2\text{CONHCl}$
- (b) $\text{CH}_3\text{CHClCOOH}$
- (c) CH_3CONH_2
- (d) $\text{CH}_3\text{CH}_2\text{COCl}$

Answer: (b)

Solution: (b) $\text{CH}_3\text{CHClCOOH}$

Question: Choose the incorrect statement about tertiary structure of protein.

Options:

- (a) Fibrous and globular are types of tertiary structures of protein
- (b) It represents the overall folding of the polypeptide chain
- (c) Disulphide linkage and hydrogen bonds stabilise the tertiary structure
- (d) During denaturation, tertiary structure remains intact

Answer: (d)

Solution: (d) During denaturation, tertiary structure remains intact

Question: Statement I: When Aluminium reacts with NaOH, $[\text{Al}(\text{OH})_6]^{3-}$ is formed.
Statement II: Shape of ClO_2^- , ClO_3^- and ClO_4^- are bent, pyramidal and tetrahedral respectively.

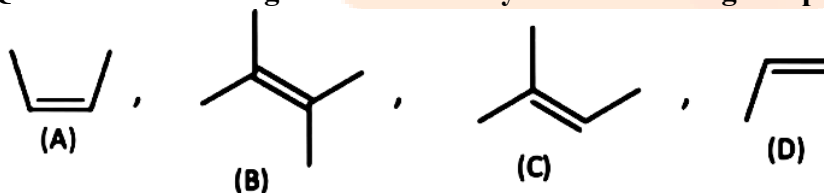
Options:

- (a) Both statement I and statement II correct
- (b) Statement I is correct and statement II is incorrect
- (c) Statement I is incorrect and statement II is correct
- (d) Both statement I and statement II are incorrect

Answer: (c)

Solution: (c) Statement I is incorrect and statement II is correct

Question: Decreasing order of stability in the following compounds is



Options:

- (a) $A > B > C > D$
- (b) $B > C > A > D$
- (c) $D > C > B > A$
- (d) $A > C > B > D$

Answer: (b)

Solution: (b) $B > C > A > D$

Question: If the bond length of molecule AB is $R_A + R_B$, while radii of A and B are R_A and R_B respectively, then which of the following relation is correct?

$(R_A > R_B)$ ($R_A > R_B$)

Options:

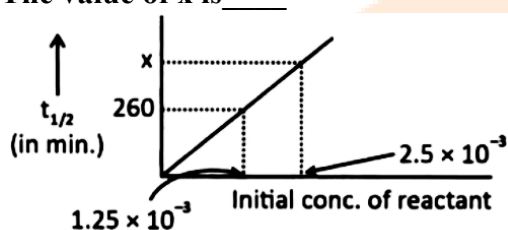
- (a) $R_{AB} = (R_A + R_B)$
- (b) $R_{AB} > (R_A + R_B)$
- (c) $R_{AB} < (R_A + R_B)$
- (d) $R_{AB} + R_A = R_B$

Answer: (c)

Solution: (c) $R_{AB} < (R_A + R_B)$

Question: For a certain reaction a graph between half life ($t_{1/2}$) and initial concentration of reactant is given

The value of x is _____



Options:

- (a) 150 min
- (b) 125 min
- (c) 520 min
- (d) 260 min

Answer: (c)

Solution:

Question: 20 g of haemoglobin is present in 1 litre solution at 300 K. The osmotic pressure was found to be 80 mm of Hg.

The molar mass of haemoglobin is (approximately):

Options:

- (a) 4424 g/mol
- (b) 4674 g/mol
- (c) 4576 g/mol
- (d) 4722 g/mol

Answer: (b)

Solution: (b) 4674 g/mol

Question: Consider the following statements:

Statement I: When value of azimuthal quantum number (l) for subshells is same, then higher the value of principal quantum number (n), higher is the energy.

Statement II: Energy of 4s subshell is greater than 3d subshell.

Options:

- (a) Both statement I and statement II are correct
- (b) Statement I is correct but statement II is incorrect
- (c) Statement I is incorrect but statement II is correct
- (d) Both statement I and statement II are incorrect

Answer: (b)

Solution: (b) Statement I is correct but statement II is incorrect

Question: Which set among the following has the same character of oxides

Options:

(a) Al_2O_3 and V_2O_3

(b) Cr_2O_3 and NO

(c) CO and N_2O

(d) SnO and N_2O_5

Answer: (c)

Solution: (c) CO and N_2O

Question: Consider the following cases in octahedral complexes:

d^3 (low spin)

d^5 (low spin)

d^7 (high spin)

d^6 (low spin)

Sum of total number of unpaired electrons present is

Options:

Answer: (7)

Solution:

Question: 20 g of pure zinc reacts with 50 mL of H_2SO_4 solution whose purity is 50%.

Density of H_2SO_4 solution is 1.3 g/mL.

The volume of H_2 gas liberated at STP (in litres) is _____.

(Consider volume of 1 mol of gas at STP = 22.4 litre)

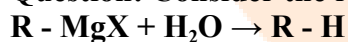
(Atomic mass: $\text{Zn} = 65$, $\text{H} = 1$, $\text{S} = 32$, $\text{O} = 16$)

Options:

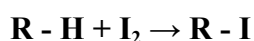
Answer: (7)

Solution:

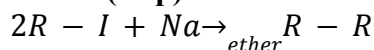
Question: Consider the reaction sequence:



1.4 dm^3/g at STP.



(1eq.)



Find molar mass of $\text{R} - \text{R}$ in g/mol.

Options:

Answer: (30)

Solution:

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[EVENING SHIFT]
Maths

Question: The coefficient of x^2 in the binomial expansion of $\left(2x^2 + \frac{1}{x}\right)^{10}$ is

Options:

- (a) 3260
- (b) 3360
- (c) 1760
- (d) 1890

Answer: (b)

Solution:

1. Write the general term

The general term T_{r+1} in the expansion of $(a + b)^n$ is given by $\binom{n}{r} a^{n-r} b^r$.

For our expression:

- $a = 2x^2$
- $b = \frac{1}{x} = x^{-1}$
- $n = 10$

$$T_{r+1} = \binom{10}{r} (2x^2)^{10-r} (x^{-1})^r$$

$$T_{r+1} = \binom{10}{r} (2^{10-r}) \cdot (x^{20-2r}) \cdot (x^{-r})$$

$$T_{r+1} = \binom{10}{r} \cdot 2^{10-r} \cdot x^{20-3r}$$

2. Solve for r

We need the coefficient of x^2 , so we set the exponent of x equal to 2:

$$20 - 3r = 2$$

$$3r = 18$$

$$r = 6$$

3. Calculate the coefficient

Substitute $r = 6$ back into the coefficient part of the general term:

$$\text{Coefficient} = \binom{10}{6} \cdot 2^{10-6}$$

$$\text{Coefficient} = \binom{10}{4} \cdot 2^4$$

First, calculate $\binom{10}{4}$:

$$\binom{10}{4} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$$

Now, multiply by 2^4 (which is 16):

$$\text{Coefficient} = 210 \times 16 = 3360$$

Question: If it is given that probability of A, becoming captain of the team is 0.8 and B becomes captain is 0.4. If A becomes captain the chance of winning the match by the team is 0.6 and when B becomes captain then chance of team winning the match is 0.7. Then the probability that team wins the match is

Options:

- (a) 0.76
- (b) 0.78
- (c) 0.74
- (d) 0.84

Answer: (a)

Solution:

1. Identify the given probabilities

From the problem description, we have two scenarios:

- **Scenario A (A is captain):**
 - Probability of A becoming captain, $P(A) = 0.8$
 - Probability of winning given A is captain, $P(W|A) = 0.6$
- **Scenario B (B is captain):**
 - Probability of B becoming captain, $P(B) = 0.4$
 - Probability of winning given B is captain, $P(W|B) = 0.7$

2. Calculate the probability of winning in each scenario

We find the joint probability of the captain being selected and the team winning for each case:

- **Winning with A:** $P(A) \times P(W|A) = 0.8 \times 0.6 = 0.48$
- **Winning with B:** $P(B) \times P(W|B) = 0.4 \times 0.7 = 0.28$

3. Determine the total probability of winning

Although the sum of $P(A)$ and $P(B)$ is greater than 1 (suggesting these might be independent matches or separate conditions considered by the coach), in this type of problem, we sum the winning probabilities from all possible paths provided:

$$\text{Total Probability of Winning} = (P(A) \times P(W|A)) + (P(B) \times P(W|B))$$

$$\text{Total Probability} = 0.48 + 0.28 = 0.76$$

Question: Let $A = \{2, 3\}$ and $B = \{5, 6\}$. Then, the number of relations from $A \times B$ to $A \times B$ are

Options:

- (a) 2^{12}
- (b) 2^{16}
- (c) 2^{10}
- (d) 2^{15}

Answer: (b)

Solution:

1. Find the number of elements in $A \times B$

- Set $A = \{2, 3\}$, so $n(A) = 2$.
- Set $B = \{5, 6\}$, so $n(B) = 2$.
- The Cartesian product $A \times B$ contains all ordered pairs (a, b) .
- $n(A \times B) = n(A) \times n(B) = 2 \times 2 = 4$.

2. Understand the definition of a Relation

A **relation** from a set X to a set Y is defined as any subset of the Cartesian product $X \times Y$.

- The number of relations is equal to the number of subsets of $X \times Y$.
- Formula: Number of relations $= 2^{n(X) \times n(Y)}$

3. Apply the formula

In this problem, both the starting set and the ending set are $A \times B$.

- Let $X = A \times B$ (where $n(X) = 4$)
- Let $Y = A \times B$ (where $n(Y) = 4$)

The total number of elements in the product set $(A \times B) \times (A \times B)$ is:

$$4 \times 4 = 16$$

Therefore, the number of possible relations is the number of subsets of these 16 elements:

$$\text{Number of relations} = 2^{16}$$

Question: A bag contains 4 red balls, 6 yellow balls and 5 blue balls. In how many ways we can select 8 balls such that we get at least two balls of each colour, is

Solution:

1. Identify the Constraints

We have:

- **Red (R):** 4 balls
- **Yellow (Y):** 6 balls
- **Blue (B):** 5 balls
- **Total to select:** 8 balls
- **Condition:** At least 2 of each color ($R \geq 2, Y \geq 2, B \geq 2$).

Let r , y , and b be the number of red, yellow, and blue balls selected. Since we must pick at least 2 of each, we first "set aside" 6 balls (2 of each color). We then only need to choose the remaining **2 balls** from any of the three colors, provided we don't exceed the total number of balls available for each color.

2. List the Possible Cases (r, y, b)

The possible combinations of counts that sum to 8 and satisfy the minimum requirements are:

1. **Case 1:** $(2, 2, 4) \rightarrow$ 2 Red, 2 Yellow, 4 Blue
2. **Case 2:** $(2, 3, 3) \rightarrow$ 2 Red, 3 Yellow, 3 Blue
3. **Case 3:** $(2, 4, 2) \rightarrow$ 2 Red, 4 Yellow, 2 Blue
4. **Case 4:** $(3, 2, 3) \rightarrow$ 3 Red, 2 Yellow, 3 Blue
5. **Case 5:** $(3, 3, 2) \rightarrow$ 3 Red, 3 Yellow, 2 Blue
6. **Case 6:** $(4, 2, 2) \rightarrow$ 4 Red, 2 Yellow, 2 Blue

3. Calculate Ways for Each Case

Assuming the balls are distinct (the standard approach for such problems), we use the

Assuming the balls are distinct (the standard approach for such problems), we use the combination formula $\binom{n}{k}$.

- **Case 1 (2,2,4):** $\binom{4}{2} \times \binom{6}{2} \times \binom{5}{4} = 6 \times 15 \times 5 = 450$
- **Case 2 (2,3,3):** $\binom{4}{2} \times \binom{6}{3} \times \binom{5}{3} = 6 \times 20 \times 10 = 1200$
- **Case 3 (2,4,2):** $\binom{4}{2} \times \binom{6}{4} \times \binom{5}{2} = 6 \times 15 \times 10 = 900$
- **Case 4 (3,2,3):** $\binom{4}{3} \times \binom{6}{2} \times \binom{5}{3} = 4 \times 15 \times 10 = 600$
- **Case 5 (3,3,2):** $\binom{4}{3} \times \binom{6}{3} \times \binom{5}{2} = 4 \times 20 \times 10 = 800$
- **Case 6 (4,2,2):** $\binom{4}{4} \times \binom{6}{2} \times \binom{5}{2} = 1 \times 15 \times 10 = 150$

4. Total Number of Ways

Now, we add the results from all cases:

$$\text{Total} = 450 + 1200 + 900 + 600 + 800 + 150 = 4100$$

Question: Let $A_1, A_2, A_3, \dots, A_{49}$ be 49 AM's between 49 and 149. Then the mean of A_1, A_{25}, A_{47} and A_{49} is

Options:

- (a) 110
- (b) 120
- (c) 130
- (d) 140

Answer: (a)

Solution:

1. Identify the common difference (d)

The sequence is 49, $A_1, A_2, \dots, A_{49}, 149$.

This is an Arithmetic Progression (AP) with $n = 49$ means inserted between $a = 49$ and $b = 149$. The total number of terms is $49 + 2 = 51$.

The common difference d is given by:

$$d = \frac{b - a}{n + 1} = \frac{149 - 49}{49 + 1} = \frac{100}{50} = 2$$

2. Calculate the specific AM's

The k -th arithmetic mean is given by the formula $A_k = a + kd$.

- $A_1 = 49 + 1(2) = 51$

- $A_{25} = 49 + 25(2) = 49 + 50 = 99$
- $A_{47} = 49 + 47(2) = 49 + 94 = 143$
- $A_{49} = 49 + 49(2) = 49 + 98 = 147$

3. Find the mean

Now, we find the average of these four values:

$$\text{Mean} = \frac{A_1 + A_{25} + A_{47} + A_{49}}{4}$$

$$\text{Mean} = \frac{51 + 99 + 143 + 147}{4}$$

$$\text{Mean} = \frac{150 + 290}{4} = \frac{440}{4} = 110$$

Question: The value of $\sum_{r=1}^{10} \frac{r}{4+r^4}$ is equal to

Options:

- (a) $\frac{4832}{123565}$
- (b) $\frac{4565}{12322}$
- (c) $\frac{4565}{12321}$
- (d) $\frac{4564}{12321}$

Answer: (b)

Solution:

Step 1: Factorize the general term

Let the general term be $T_r = \frac{r}{r^4 + 4}$. Using **Sophie Germain's Identity**, we can factorize the denominator:

$$r^4 + 4 = (r^2 + 2)^2 - (2r)^2$$

$$r^4 + 4 = (r^2 + 2r + 2)(r^2 - 2r + 2)$$

Now, notice that the difference between these two factors is:

$$(r^2 + 2r + 2) - (r^2 - 2r + 2) = 4r$$

We can rewrite T_r as:

$$T_r = \frac{1}{4} \left[\frac{4r}{(r^2 - 2r + 2)(r^2 + 2r + 2)} \right]$$

$$T_r = \frac{1}{4} \left[\frac{(r^2 + 2r + 2) - (r^2 - 2r + 2)}{(r^2 - 2r + 2)(r^2 + 2r + 2)} \right]$$

$$T_r = \frac{1}{4} \left[\frac{1}{r^2 - 2r + 2} - \frac{1}{r^2 + 2r + 2} \right]$$

Step 2: Set up the telescoping series

Observe that $r^2 - 2r + 2 = (r - 1)^2 + 1$ and $r^2 + 2r + 2 = (r + 1)^2 + 1$.

Let $f(k) = \frac{1}{k^2 + 1}$. Then the general term becomes:

$$T_r = \frac{1}{4} [f(r - 1) - f(r + 1)]$$

Summing from $r = 1$ to 10:

$$\text{Sum} = \frac{1}{4} \sum_{r=1}^{10} [f(r - 1) - f(r + 1)]$$

$$\text{Sum} = \frac{1}{4} [(f(0) - f(2)) + (f(1) - f(3)) + (f(2) - f(4)) + \dots + (f(9) - f(11))]$$

Most terms cancel out, leaving:

$$\text{Sum} = \frac{1}{4} [f(0) + f(1) - f(10) - f(11)]$$

Step 3: Calculate the final value

Using $f(k) = \frac{1}{k^2 + 1}$:

- $f(0) = \frac{1}{0^2 + 1} = 1$
- $f(1) = \frac{1}{1^2 + 1} = \frac{1}{2}$
- $f(10) = \frac{1}{10^2 + 1} = \frac{1}{101}$
- $f(11) = \frac{1}{11^2 + 1} = \frac{1}{122}$

Substitute these values back:

$$\begin{aligned}\text{Sum} &= \frac{1}{4} \left(1 + \frac{1}{2} - \frac{1}{101} - \frac{1}{122} \right) \\ \text{Sum} &= \frac{1}{4} \left(\frac{3}{2} - \frac{122 + 101}{101 \times 122} \right) = \frac{1}{4} \left(\frac{3}{2} - \frac{223}{12322} \right) \\ \text{Sum} &= \frac{1}{4} \left(\frac{3 \times 6161 - 223}{12322} \right) = \frac{1}{4} \left(\frac{18483 - 223}{12322} \right) \\ \text{Sum} &= \frac{1}{4} \times \frac{18260}{12322} = \frac{4565}{12322}\end{aligned}$$

Question: If α, β are the roots of the equation $x^2 - 4x + p = 0$ and γ and δ are the roots of the equation $x^2 - x + q = 0$. When $\alpha, \beta, \gamma, \delta$ form a GP with positive common ratio, then the value of $p + q$ is

Options:

- (a) $\frac{22}{9}$
- (b) $\frac{33}{9}$
- (c) $\frac{21}{9}$
- (d) $\frac{34}{9}$

Answer: (d)

Solution:

1. Identify Relations from the Equations

For a quadratic equation $ax^2 + bx + c = 0$, the sum of roots is $-b/a$ and the product of roots is c/a .

- **Equation 1:** $x^2 - 4x + p = 0$ with roots α, β
 - $\alpha + \beta = 4$
 - $\alpha\beta = p$
- **Equation 2:** $x^2 - x + q = 0$ with roots γ, δ
 - $\gamma + \delta = 1$
 - $\gamma\delta = q$

2. Apply the Geometric Progression (GP) Condition

Since $\alpha, \beta, \gamma, \delta$ form a GP, we can represent them using the first term a and a common ratio r :

- $\alpha = a$
- $\beta = ar$
- $\gamma = ar^2$
- $\delta = ar^3$

Now, substitute these into the sum equations we found in Step 1:

$$1. a + ar = 4 \implies a(1 + r) = 4$$

$$2. ar^2 + ar^3 = 1 \implies ar^2(1 + r) = 1$$

3. Solve for r and a

Divide the second equation by the first:

$$\frac{ar^2(1 + r)}{a(1 + r)} = \frac{1}{4}$$

$$r^2 = \frac{1}{4}$$

Since the problem states that the common ratio is **positive**, we have:

$$r = \frac{1}{2}$$

Now, substitute $r = 1/2$ back into the first sum equation to find a :

$$a \left(1 + \frac{1}{2} \right) = 4$$

$$a \left(\frac{3}{2} \right) = 4 \implies a = \frac{8}{3}$$

4. Find the values of p and q

Using the product of roots:

- For p :

$$p = \alpha\beta = a \cdot ar = a^2r$$

$$p = \left(\frac{8}{3}\right)^2 \cdot \frac{1}{2} = \frac{64}{9} \cdot \frac{1}{2} = \frac{32}{9}$$

- For q :

$$q = \gamma\delta = ar^2 \cdot ar^3 = a^2r^5$$

$$q = \left(\frac{8}{3}\right)^2 \cdot \left(\frac{1}{2}\right)^5 = \frac{64}{9} \cdot \frac{1}{32} = \frac{2}{9}$$

5. Final Calculation

The question asks for the value of $p + q$:

$$p + q = \frac{32}{9} + \frac{2}{9} = \frac{34}{9}$$

Question: If Z_1 and Z_2 are the solutions of the equation $6Z^2 - 11Z + 59 - 3i = 0$ then

$|Z_1|^2 + |Z_2|^2$ is equal to

Options:

- (a) $\frac{709}{36}$
- (b) $\frac{703}{18}$
- (c) $\frac{705}{36}$
- (d) $\frac{701}{36}$

Answer: (a)

Solution:

1. Identify the Coefficients

The given equation is $6z^2 - 11z + (59 - 3i) = 0$.
Comparing this with $az^2 + bz + c = 0$, we have:

- $a = 6$
- $b = -11$
- $c = 59 - 3i$

2. Find the Discriminant (D)

Using the formula $D = b^2 - 4ac$:

$$D = (-11)^2 - 4(6)(59 - 3i)$$

$$D = 121 - 24(59 - 3i)$$

$$D = 121 - 1416 + 72i$$

$$D = -1295 + 72i$$

3. Find the Square Root of D

Let $\sqrt{-1295 + 72i} = x + iy$. Squaring both sides:
 $x^2 - y^2 = -1295$ and $2xy = 72 \implies xy = 36$

Using the property $|x + iy|^2 = \sqrt{(-1295)^2 + 72^2}$:

$$x^2 + y^2 = \sqrt{1677025 + 5184} = \sqrt{1682209} = 1297$$

Solving the system:

- $2x^2 = 1297 - 1295 = 2 \implies x = \pm 1$
 - $2y^2 = 1297 + 1295 = 2592 \implies y = \pm 36$
- Since xy is positive, x and y have the same sign. So, $\sqrt{D} = \pm (1 + 36i)$.

4. Solve for z_1 and z_2

Using the quadratic formula $z = \frac{-b \pm \sqrt{D}}{2a}$:

$$z = \frac{11 \pm (1 + 36i)}{12}$$

- **Root 1 (z_1):** $\frac{11 + 1 + 36i}{12} = \frac{12 + 36i}{12} = 1 + 3i$
- **Root 2 (z_2):** $\frac{11 - 1 - 36i}{12} = \frac{10 - 36i}{12} = \frac{5}{6} - 3i$

5. Calculate $|z_1|^2 + |z_2|^2$

Recall that $|x + iy|^2 = x^2 + y^2$.

- $|z_1|^2 = 1^2 + 3^2 = 1 + 9 = 10$
- $|z_2|^2 = \left(\frac{5}{6}\right)^2 + (-3)^2 = \frac{25}{36} + 9 = \frac{25 + 324}{36} = \frac{349}{36}$

Total Sum:

$$10 + \frac{349}{36} = \frac{360 + 349}{36} = \frac{709}{36}$$

Question: The number of solution of equation $\cos \theta \cos \frac{5\theta}{2} = \cos 7\theta \cos \frac{7\theta}{2}$, $\theta \in [-\pi, \pi]$ is equal to

Solution:

1. Apply the Product-to-Sum Identity

Recall the identity: $2\cos A \cos B = \cos(A + B) + \cos(A - B)$.

Multiplying both sides of the given equation by 2, we get:

$$2\cos \theta \cos \frac{5\theta}{2} = 2\cos 7\theta \cos \frac{7\theta}{2}$$

Applying the identity:

$$\cos\left(\theta + \frac{5\theta}{2}\right) + \cos\left(\theta - \frac{5\theta}{2}\right) = \cos\left(7\theta + \frac{7\theta}{2}\right) + \cos\left(7\theta - \frac{7\theta}{2}\right)$$

$$\cos\left(\frac{7\theta}{2}\right) + \cos\left(-\frac{3\theta}{2}\right) = \cos\left(\frac{21\theta}{2}\right) + \cos\left(\frac{7\theta}{2}\right)$$

2. Simplify the Equation

Notice that $\cos\left(\frac{7\theta}{2}\right)$ appears on both sides, so they cancel out. Also, since $\cos(-x) = \cos$

(x):

$$\cos\left(\frac{3\theta}{2}\right) = \cos\left(\frac{21\theta}{2}\right)$$

3. Solve for General Solutions

The general solution for $\cos X = \cos Y$ is $X = 2n\pi \pm Y$, where n is an integer.

$$\frac{21\theta}{2} = 2n\pi \pm \frac{3\theta}{2}$$

Case 1: Plus sign (+)

$$\frac{21\theta}{2} = 2n\pi + \frac{3\theta}{2} \implies \frac{18\theta}{2} = 2n\pi \implies 9\theta = 2n\pi \implies \theta = \frac{2n\pi}{9}$$

For $\theta \in [-\pi, \pi]$, n can be $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. (9 solutions)

Case 2: Minus sign (-)

$$\frac{21\theta}{2} = 2n\pi - \frac{3\theta}{2} \implies \frac{24\theta}{2} = 2n\pi \implies 12\theta = 2n\pi \implies \theta = \frac{n\pi}{6}$$

For $\theta \in [-\pi, \pi]$, n can be $\{-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$. (13 solutions)

4. Find Unique Solutions

We must subtract the solutions that are common to both cases to avoid double-counting.

Common values occur when $\frac{2n_1\pi}{9} = \frac{n_2\pi}{6}$, which simplifies to $4n_1 = 3n_2$.

- If $n_1 = 0$, then $n_2 = 0 \implies \theta = 0$
- If $n_1 = 3$, then $n_2 = 4 \implies \theta = \frac{2\pi}{3}$
- If $n_1 = -3$, then $n_2 = -4 \implies \theta = -\frac{2\pi}{3}$

There are 3 common solutions.

Total unique solutions = $9 + 13 - 3 = 19$.

The number of solutions in the interval $[-\pi, \pi]$ is 19.

Question: If $y = y(x)$ be the solution of the differential equation

$$(\sqrt{\tan x}) dy = \left(\sec^3 x - (\tan x)^{\frac{3}{2}} y \right) dx \text{ and } y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5} \text{ and } y\left(\frac{\pi}{3}\right) = \frac{4\alpha}{5} \text{ then } \alpha^4 \text{ is}$$

equal to

Solution:

1. Standard Form of the Equation

First, let's rearrange the given equation into the standard linear form:

$$\frac{dy}{dx} + P(x)y = Q(x).$$

Dividing the original equation by $(\sqrt{\tan x})dx$:

$$\frac{dy}{dx} = \frac{\sec^3 x}{\sqrt{\tan x}} - \frac{(\tan x)^{3/2}}{\sqrt{\tan x}} y$$

$$\frac{dy}{dx} + (\tan x)y = \frac{\sec^3 x}{\sqrt{\tan x}}$$

2. Find the Integrating Factor (I.F.)

The integrating factor is $e^{\int P(x)dx}$:

$$I.F. = e^{\int \tan x dx} = e^{\ln(\sec x)} = \sec x$$

3. Integrate to Find the General Solution

The solution is given by $y \cdot (I.F.) = \int Q(x) \cdot (I.F.) dx$:

$$y \cdot \sec x = \int \frac{\sec^3 x}{\sqrt{\tan x}} \cdot \sec x dx = \int \frac{\sec^4 x}{\sqrt{\tan x}} dx$$

Using $\sec^2 x = 1 + \tan^2 x$, the integral becomes:

$$y \sec x = \int \frac{(1 + \tan^2 x) \sec^2 x}{\sqrt{\tan x}} dx$$

Let $u = \tan x \implies du = \sec^2 x dx$:

$$y \sec x = \int (u^{-1/2} + u^{3/2}) du = 2u^{1/2} + \frac{2}{5} u^{5/2} + C$$

$$y \sec x = 2\sqrt{\tan x} + \frac{2}{5} (\tan x)^{5/2} + C$$

4. Determine the Constant C

Using the condition $y(\frac{\pi}{4}) = \frac{6\sqrt{2}}{5}$ (where $\tan \frac{\pi}{4} = 1$ and $\sec \frac{\pi}{4} = \sqrt{2}$):

$$\frac{6\sqrt{2}}{5} \cdot \sqrt{2} = 2(1) + \frac{2}{5}(1) + C$$

$$\frac{12}{5} = \frac{10+2}{5} + C \Rightarrow C = 0$$

5. Calculate α^4

Now, use the condition $y(\frac{\pi}{3}) = \frac{4\alpha}{5}$ (where $\tan \frac{\pi}{3} = \sqrt{3}$ and $\sec \frac{\pi}{3} = 2$):

$$\frac{4\alpha}{5} \cdot 2 = 2(\sqrt{3})^{1/2} + \frac{2}{5}(\sqrt{3})^{5/2}$$

$$\frac{8\alpha}{5} = 2(3^{1/4}) + \frac{2 \cdot 3^2 \cdot 3^{1/2}}{5} \text{ (Wait, } (\sqrt{3})^{5/2} = 3^{5/4} = 3 \cdot 3^{1/4})$$

$$\frac{8\alpha}{5} = 2(3^{1/4}) + \frac{6}{5}(3^{1/4})$$

$$\frac{8\alpha}{5} = \frac{10 \cdot 3^{1/4} + 6 \cdot 3^{1/4}}{5} = \frac{16 \cdot 3^{1/4}}{5}$$

$$8\alpha = 16 \cdot 3^{1/4} \Rightarrow \alpha = 2 \cdot 3^{1/4}$$

Finally, raising to the fourth power:

$$\alpha^4 = (2 \cdot 3^{1/4})^4 = 16 \cdot 3 = 48$$

Question: If the Mean of the frequency distribution table

X_i	0	2	6	12			$n(n+1)$
f_i	nC_0	nC_1	nC_2	nC_3			${}^nC_\square$

is 45. Then the median will be

Solution:

1. Find the Value of n using the Mean

From the table, we can see the general terms for the values X_i and frequencies f_i :

- $X_i = i(i + 1)$ for $i = 0, 1, 2, \dots, n$.
- $f_i = {}^nC_i$

The formula for the mean ($\bar{x}$) is:

$$\bar{x} = \frac{\sum_{i=0}^n f_i X_i}{\sum_{i=0}^n f_i} = \frac{\sum_{i=0}^n {}^nC_i \cdot i(i + 1)}{\sum_{i=0}^n {}^nC_i}$$

Numerator:

Using the identities $\sum i \cdot {}^nC_i = n \cdot 2^{n-1}$ and $\sum i^2 \cdot {}^nC_i = n(n + 1)2^{n-2}$, we get:

$$\sum i(i + 1) \cdot {}^nC_i = \sum (i^2 + i) \cdot {}^nC_i = n(n + 1)2^{n-2} + n \cdot 2^{n-1}$$

Factoring this out simplifies to: $n(n + 3)2^{n-2}$.

Denominator:

The sum of binomial coefficients is $\sum {}^nC_i = 2^n$.

Calculating n :

$$\bar{x} = \frac{n(n + 3)2^{n-2}}{2^n} = \frac{n(n + 3)}{4}$$

Given the mean is 45:

$$\frac{n(n + 3)}{4} = 45 \implies n(n + 3) = 180$$

Solving the quadratic $n^2 + 3n - 180 = 0 \implies (n + 15)(n - 12) = 0$. Since n must be positive, $n = 12$.

2. Find the Median

For $n = 12$, the frequencies are the binomial coefficients ${}^{12}C_0, {}^{12}C_1, \dots, {}^{12}C_{12}$.

The total frequency is $N = 2^{12} = 4096$. The median is the value at the $\frac{N}{2} = 2048^{th}$

The total frequency is $N = 2^{12} = 4096$. The median is the value at the $\frac{N}{2} = 2048^{th}$ position.

Because binomial coefficients are **symmetric**, the middle term occurs at $i = \frac{n}{2} = 6$.

- Sum of frequencies up to $i = 5$: ≈ 1586
- Sum of frequencies up to $i = 6$: ≈ 2510

Since the 2048^{th} position falls within the frequency of the term where $i = 6$, the median value is X_6 .

$$X_6 = 6(6 + 1) = 6 \cdot 7 = 42$$

Question: Let the eccentricity of a hyperbola satisfy the equation $6e^2 - 11e + 3 = 0$. If the foci of the hyperbola are $(3, 5)$ and $(3, -4)$, then the length of its latus rectum is:

Solution:

1. Find the Eccentricity (e)

We are given the quadratic equation for the eccentricity:

$$6e^2 - 11e + 3 = 0$$

Let's factorize it:

$$6e^2 - 9e - 2e + 3 = 0$$

$$3e(2e - 3) - 1(2e - 3) = 0$$

$$(3e - 1)(2e - 3) = 0$$

This gives two possible values: $e = \frac{1}{3}$ or $e = \frac{3}{2}$.

Since we are dealing with a **hyperbola**, the eccentricity must be greater than 1 ($e > 1$). Therefore, we take:

$$e = \frac{3}{2}$$

2. Determine the Distance Between Foci ($2ae$)

The foci are given as $(3, 5)$ and $(3, -4)$. Since the x -coordinates are the same, the hyperbola is vertical. The distance between the foci is the difference in their y -coordinates:

$$2ae = |5 - (-4)| = 9$$

Now, substitute $e = \frac{3}{2}$ to find a :

$$2 \cdot a \cdot \frac{3}{2} = 9$$

$$3a = 9 \implies a = 3$$

3. Calculate b^2

For a hyperbola, the relationship between a , b , and e is:

$$b^2 = a^2(e^2 - 1)$$

Substitute the values we found:

$$b^2 = 3^2 \left(\left(\frac{3}{2} \right)^2 - 1 \right)$$

$$b^2 = 9 \left(\frac{9}{4} - 1 \right) = 9 \left(\frac{5}{4} \right) = \frac{45}{4}$$

4. Find the Length of the Latus Rectum

The formula for the length of the latus rectum of a hyperbola is:

$$L.R. = \frac{2b^2}{a}$$

Plugging in our values:

$$L.R. = \frac{2 \cdot \left(\frac{45}{4} \right)}{3}$$

$$L.R. = \frac{\frac{45}{2}}{3} = \frac{45}{6} = 7.5 \text{ or } \frac{15}{2}$$

Question: Let $f(x)$ and $g(x)$ be the twice differentiable functions satisfying $f''(x) = g''(x)$, for all $x \in \mathbb{R}$, such that $f'(1) = 2g'(1) = 4$ and $g(2) = 2f(2) = 9$. Then $f(25) - g(25)$ equal to:

Solution:

1. Define the Difference Function

Let $h(x) = f(x) - g(x)$.

Since $f(x)$ and $g(x)$ are twice differentiable, $h(x)$ is also twice differentiable.

From the problem, we know that $f''(x) = g''(x)$ for all x . This means:

$$h''(x) = f''(x) - g''(x) = 0$$

If the second derivative of a function is zero everywhere, then its first derivative must be a **constant**.

$$h'(x) = f'(x) - g'(x) = c_1$$

2. Find the Constant c_1

We are given information about the derivatives at $x = 1$:

- $f'(1) = 4$

- $2g'(1) = 4 \implies g'(1) = 2$

Now, substitute these into our expression for $h'(x)$:

$$h'(1) = f'(1) - g'(1) = 4 - 2 = 2$$

Since $h'(x)$ is a constant, $h'(x) = 2$ for all x .

3. Find the Value of $h(25)$

We can use the **Fundamental Theorem of Calculus** to relate the values of $h(x)$ at $x = 2$ and $x = 25$:

$$h(25) - h(2) = \int_2^{25} h'(x) dx$$

First, let's calculate $h(2)$ using the given values:

- $g(2) = 9$
- $2f(2) = 9 \implies f(2) = 4.5$
- $h(2) = f(2) - g(2) = 4.5 - 9 = -4.5$

Now, perform the integration:

$$h(25) - (-4.5) = \int_2^{25} 2 dx$$

$$h(25) + 4.5 = [2x]_2^{25}$$

$$h(25) + 4.5 = 2(25) - 2(2) = 50 - 4 = 46$$

4. Final Calculation

Solve for $h(25)$:

$$h(25) = 46 - 4.5 = 41.5$$

Therefore, $f(25) - g(25) = 41.5$.

Question: Let $f(x) = \lim_{y \rightarrow 0} \frac{(1 - \cos(xy)) \tan(xy)}{y^3}$. Then number of solutions of the equation $f(x) = \sin x$, $x \in \mathbb{R}$.

Solution:

1. Evaluate the Limit for $f(x)$

The limit is taken as $y \rightarrow 0$, treating x as a constant. To simplify the expression, we can use standard trigonometric limits. Let's rewrite the expression to match those standard forms:

$$f(x) = \lim_{y \rightarrow 0} \left[\frac{1 - \cos(xy)}{(xy)^2} \cdot \frac{\tan(xy)}{xy} \cdot \frac{(xy)^3}{y^3} \right]$$

Since $xy \rightarrow 0$ as $y \rightarrow 0$, we apply the known limits:

- $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}$
- $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$

Substituting these back into the expression:

$$f(x) = \frac{1}{2} \cdot 1 \cdot \frac{x^3 y^3}{y^3}$$

$$f(x) = \frac{x^3}{2}$$

2. Set Up the Equation

The problem asks for the number of solutions to $f(x) = \sin x$, which is now:

$$\frac{x^3}{2} = \sin x \implies x^3 = 2 \sin x$$

3. Analyze the Number of Solutions

Let $g(x) = x^3 - 2\sin x$. We are looking for the roots of $g(x) = 0$.

- **Trivial Solution:** Since $g(0) = 0^3 - 2\sin(0) = 0$, $x = 0$ is one solution.
- **Symmetry:** Both x^3 and $\sin x$ are **odd functions**, meaning $g(x)$ is also an odd function. This tells us that if $x = \alpha$ is a solution, then $x = -\alpha$ must also be a solution.
- **Positive Roots:** Let's look at the behavior for $x > 0$.
 - The second derivative is $g''(x) = 6x + 2\sin x$. For $x > 0$, $g''(x)$ is always positive (since $6x$ grows much faster than $2\sin x$), making the function **strictly convex** for $x > 0$.

- The first derivative is $g'(x) = 3x^2 - 2\cos x$. At $x = 0$, $g'(0) = -2$, meaning the function starts by decreasing into the negative region.
- As $x \rightarrow \infty$, $g(x) \rightarrow \infty$.
- Because the function starts at 0, decreases, and is concave up (convex), it must cross the x-axis exactly **once** at some positive value.
- Checking values: $g(1) \approx -0.68$ and $g(\pi/2) \approx 1.87$. So, there is a root between 1 and 1.57.

4. Conclusion

We have found:

1. One solution at $x = 0$.
2. One unique solution for $x > 0$.
3. One unique solution for $x < 0$ (due to symmetry).

Total number of solutions = 3.

Question: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x$, $x \in \mathbb{R}$. Let the maximum value of f on $\mathbb{R}$ be α . If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3$, $\beta > 0$ is α^2 , then $30\beta^2$ is equal to ____.

Solution:

1. Determine the Function $f(x)$

We are given the functional equation:

$$1. f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x$$

To find $f(x)$, we replace x with $\left(\frac{\pi}{2} - x\right)$:

$$f\left(\frac{\pi}{2} - x\right) + 3f\left(\frac{\pi}{2} - \left(\frac{\pi}{2} - x\right)\right) = \sin\left(\frac{\pi}{2} - x\right)$$

$$2. f\left(\frac{\pi}{2} - x\right) + 3f(x) = \cos x$$

Now, we have a system of two linear equations. Multiply the second equation by 3:

$$3f\left(\frac{\pi}{2} - x\right) + 9f(x) = 3\cos x$$

Subtract the first equation from this new equation to eliminate $f\left(\frac{\pi}{2} - x\right)$:

$$8f(x) = 3\cos x - \sin x$$

$$f(x) = \frac{3\cos x - \sin x}{8}$$

2. Find the Maximum Value α

The maximum value of an expression $A\cos x + B\sin x$ is $\sqrt{A^2 + B^2}$.

For the numerator $3\cos x - \sin x$, the maximum value is $\sqrt{3^2 + (-1)^2} = \sqrt{10}$.

Thus, the maximum value of $f(x)$ is:

$$\alpha = \frac{\sqrt{10}}{8}$$

Squaring this for later use in the area equation:

$$\alpha^2 = \frac{10}{64} = \frac{5}{32}$$

3. Calculate the Bounded Area

The curves are $g(x) = x^2$ and $h(x) = \beta x^3$ (where s in the image is a typo for x). To find the area, first find the intersection points:

$$x^2 = \beta x^3 \implies x^2(1 - \beta x) = 0$$

The curves intersect at $x = 0$ and $x = \frac{1}{\beta}$.

The area A is given by the integral:

$$A = \int_0^{1/\beta} (x^2 - \beta x^3) dx$$

$$A = \left[\frac{x^3}{3} - \frac{\beta x^4}{4} \right]_0^{1/\beta} = \frac{1}{3\beta^3} - \frac{\beta}{4\beta^4} = \frac{1}{3\beta^3} - \frac{1}{4\beta^3} = \frac{1}{12\beta^3}$$

4. Solve for $30\beta^3$

We are told the area is equal to α^2 :

$$\frac{1}{12\beta^3} = \frac{5}{32}$$

Cross-multiplying gives:

$$60\beta^3 = 32$$

Divide both sides by 2 to find the requested value:

$$30\beta^3 = 16$$

Question: If $f(x)$ satisfy the equation $f(x) = \int_1^x f(t)dt + (1 - x)(\log_e x - 1) + e$ then

$f(f(1))$ is equal to

Options:

(a) $1 - e^{2-e}$

(b) $1 + e^{2+e}$

(c) $1 + e^e$

(d) $1 - e^{2+e}$

Answer: (c)

Solution:

1. Find the value of $f(1)$

First, we substitute $x = 1$ into the original equation to find the initial condition:

$$f(1) = \int_1^1 f(t)dt + (1-1)(\log_e 1-1) + e$$

Since the integral from 1 to 1 is 0 and $(1-1) = 0$:

$$f(1) = 0 + 0 + e = e$$

2. Formulate the Differential Equation

To remove the integral, we differentiate both sides of the equation with respect to x using the **Leibniz Rule**:

$$\frac{d}{dx} f(x) = \frac{d}{dx} \left[\int_1^x f(t)dt \right] + \frac{d}{dx} [(1-x)(\ln x - 1)] + \frac{d}{dx} [e]$$

- The derivative of the integral is $f(x)$.
- Using the product rule for the second term:

$$\frac{d}{dx} [(1-x)(\ln x - 1)] = (-1)(\ln x - 1) + (1-x)\left(\frac{1}{x}\right) = -\ln x + 1 + \frac{1}{x} - 1 = \frac{1}{x} - \ln x$$

This gives us the first-order linear differential equation:

$$f'(x) = f(x) + \frac{1}{x} - \ln x \implies f'(x) - f(x) = \frac{1}{x} - \ln x$$

3. Solve the Differential Equation

We use the **Integrating Factor (I.F.)** method:

$$I.F. = e^{\int -1dx} = e^{-x}$$

Multiply the entire equation by e^{-x} :

$$e^{-x} \cdot f'(x) - e^{-x} \cdot f(x) = e^{-x} \left(\frac{1}{x} - \ln x \right)$$

$$\frac{d}{dx} [f(x) \cdot e^{-x}] = e^{-x} \left(\frac{1}{x} - \ln x \right)$$

Integrating both sides:

$$f(x) \cdot e^{-x} = \int e^{-x} \left(\frac{1}{x} - \ln x \right) dx$$

Notice that the derivative of $(e^{-x} \ln x)$ is exactly $e^{-x}(\frac{1}{x}) - e^{-x} \ln x$. Thus:

$$f(x) \cdot e^{-x} = e^{-x} \ln x + C$$

$$f(x) = \ln x + Ce^x$$

4. Find the Constant and Calculate $f(f(1))$

Using our initial condition $f(1) = e$:

$$e = \ln(1) + Ce^1 \implies e = 0 + Ce \implies C = 1$$

So, the function is $f(x) = \ln x + e^x$. Now, we calculate $f(f(1))$:

$$f(f(1)) = f(e)$$

$$f(e) = \ln(e) + e^e = 1 + e^e$$

Question: Let position vector of two points P and Q are $\vec{a}$ and $\vec{b}$ respectively. If R be a point on OP such that $\vec{OR} = \frac{1}{5}\vec{OP}$ and M be another point such that $\vec{QM} = \frac{1}{5}\vec{OM}$, then $\vec{RM}$ is equal to (where O is origin)

Options:

(a) $\frac{4\vec{b}}{5} - \frac{\vec{a}}{5}$

(b) $\frac{5\vec{b}}{4} - \frac{\vec{a}}{5}$

(c) $\frac{\vec{b}}{4} - \frac{\vec{a}}{5}$

(d) $\frac{4\vec{b}}{5} - \frac{\vec{a}}{4}$

Answer: (b)

Solution:

1. Identify the given position vectors

- Position vector of P : $\vec{OP} = \vec{a}$
- Position vector of Q : $\vec{OQ} = \vec{b}$

2. Express $\vec{OR}$ in terms of $\vec{a}$

We are given the relation $5\vec{OR} = \vec{OP}$. Since $\vec{OP} = \vec{a}$:

$$\vec{OR} = \frac{\vec{a}}{5}$$

3. Express $\vec{OM}$ in terms of $\vec{b}$

We are given $5\vec{QM} = \vec{OM}$. To solve for $\vec{OM}$, we use the property that $\vec{QM} = \vec{OM} - \vec{OQ}$:

$$5(\vec{OM} - \vec{OQ}) = \vec{OM}$$

$$5\vec{OM} - 5\vec{b} = \vec{OM}$$

$$4\vec{OM} = 5\vec{b}$$

$$\vec{OM} = \frac{5\vec{b}}{4}$$

4. Find the vector $\vec{RM}$

The vector $\vec{RM}$ is calculated by subtracting the position vector of the starting point from the ending point:

$$\vec{RM} = \vec{OM} - \vec{OR}$$

Substituting the values we found:

$$\vec{RM} = \frac{5\vec{b}}{4} - \frac{\vec{a}}{5}$$